# AFRICAN GIG-ECONOMY & DIGITAL WALLET RISK

## Analytical Report & Data Quality Audit

Scope: 50,000 transactions · 5,000 workers · 4 markets · 12 channels · Jan 2023 – Dec 2024

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Report date: August 18, 2026

⚠️ *Headline finding: the risk patterns hypothesized in the analysis brief are not statistically present in this dataset. See Section 3 for full verification.*

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# 1. Executive Summary

This report analyzes 50,000 digital-wallet and gig-economy transactions across four African markets (Nigeria, Kenya, Ghana, South Africa) spanning January 2023 to December 2024, covering 5,000 gig workers transacting through 12 channel configurations.

## 1.1 The headline finding

The analysis brief supplied with this dataset proposed six specific risk patterns (e.g., "USSD channel shows 2.3x higher fraud rate than app channel"). We tested every one of them directly against the data using chi-square tests of independence and Pearson correlations. None of the six held up. Fraud, dispute, and reversal flags are statistically indistinguishable from a uniform 50% coin-flip across every channel, country, worker segment, tenure band, and velocity level we tested — and every claimed relationship is well inside normal sampling noise (all p-values > 0.05, most > 0.4).

**This is the single most important thing for stakeholders to know before acting on this dataset: it behaves like independently, randomly assigned risk labels rather than production fraud data with real underlying drivers. Section 3 documents the full verification, one claim at a time, so this conclusion is auditable rather than asserted.**

## 1.2 What is real and usable in this dataset

- Total transaction value: $24,975,394 across 50,000 transactions (avg $499.51/transaction, remarkably uniform — see Section 2.2).
- The star schema itself is clean: zero missing values in the fact table, zero duplicate transaction IDs, and 100% referential integrity across all four dimension tables (every worker_id, channel_id, market_id, and date_id in the fact table resolves to a dimension row).
- One real data-quality defect was found and fixed: the week_number column in dim_date was 100% null in the source file. It has been recalculated as a true ISO 8601 week number directly in the Power BI file's Power Query step, so the fix survives a full data refresh rather than living only in a static export.
- Base-rate risk exposure is genuinely large in absolute terms — $12.56M in flagged fraud loss (50.3% of transactions flagged) — even though it isn't concentrated anywhere. That has its own operational implication (Section 4).

## 1.3 Recommendations at a glance

- Do not build channel-, country-, tenure-, or segment-based risk rules from this dataset — none of those splits carry signal here. Rules built on this data would fire close to 50% of the time regardless of the input feature, which is operationally equivalent to no rule at all.
- Treat this as a synthetic / template dataset, not a production extract, and request (or generate) a dataset where labels are actually driven by the input features before doing model development or writing risk policy from it.
- If this data is a fixture for testing a fraud pipeline or dashboard, that's a legitimate use — the pipeline and dashboard logic built here will work correctly, it's only the substantive conclusions that don't transfer to a real portfolio.
- Where a real extract becomes available, re-run Section 3's exact test battery unchanged — it is designed as a repeatable validation harness, not a one-off critique of this file.

# 2. Data Overview & Quality Audit

## 2.1 Schema

The dataset is a standard star schema: one fact table (fact_transactions, 50,000 rows, 15 columns) and four dimension tables.

| Table | Rows | Grain / Key | Notes |
|---|---|---|---|
| dim_worker | 5,000 | 1 row per gig worker | gig_segment, kyc_tier, risk_score, tenure, demographics |
| dim_channel | 12 | 1 row per channel config | channel_type × channel_subtype combinations |
| dim_market | 4 | 1 row per country | Nigeria, Kenya, Ghana, South Africa |
| dim_date | 731 | 1 row per calendar day | 2023-01-01 to 2024-12-31 inclusive |
| fact_transactions | 50,000 | 1 row per transaction | 13 transaction types, 8 outcome states |

## 2.2 Data quality findings

- Completeness: zero nulls across all 15 fact-table columns and all dimension columns except the week_number defect noted below.
- Uniqueness: 50,000/50,000 transaction_ids are unique; no duplicate rows.
- Referential integrity: 100% of fact-table foreign keys (worker_id, channel_id, market_id, date_id) resolve to an existing dimension row — a clean join with zero orphans.
- Defect — dim_date.week_number: 100% null in the source file (731/731 rows). Fixed directly in Power Query with a custom column implementing the ISO 8601 week algorithm (shift to the Thursday of the week, count days since that Thursday's January 1, divide by 7, round up), so the value regenerates correctly on every refresh rather than needing to be reapplied by hand.
- Notable synthetic-data signature: amount_usd, velocity_score, risk_score, and the dim_channel/dim_market "fraud_rate"/"fraud_index" fields are all near-perfectly uniform distributions over their range (e.g., amount_usd spans $0.02–$999.97 with mean $499.51, almost exactly the midpoint) — a pattern consistent with random generation (Uniform(0,1000)) rather than an observed real-world long-tailed transaction-value distribution.
- fraud_loss_usd is mechanically derived from is_fraud_flagged, not an independent signal: it is exactly $0 for every one of the 24,855 non-flagged transactions and strictly positive for every one of the 25,145 flagged transactions. It is a consequence of the flag, not a predictor of it.

## 2.3 Volume & value snapshot

| Metric | Value |
|---|---|
| Total transactions | 50,000 |
| Total value processed | $24,975,393.59 |

| | |
|---|---|
| Average transaction value | $499.51 |
| Fraud-flagged rate | 50.29% |
| Disputed rate | 50.07% |
| Reversed rate | 49.90% |
| Total flagged fraud loss | $12,559,256.91 |
| Avg loss per flagged case | $499.47 |
| Active workers | 2,441 / 5,000 (48.8%) |

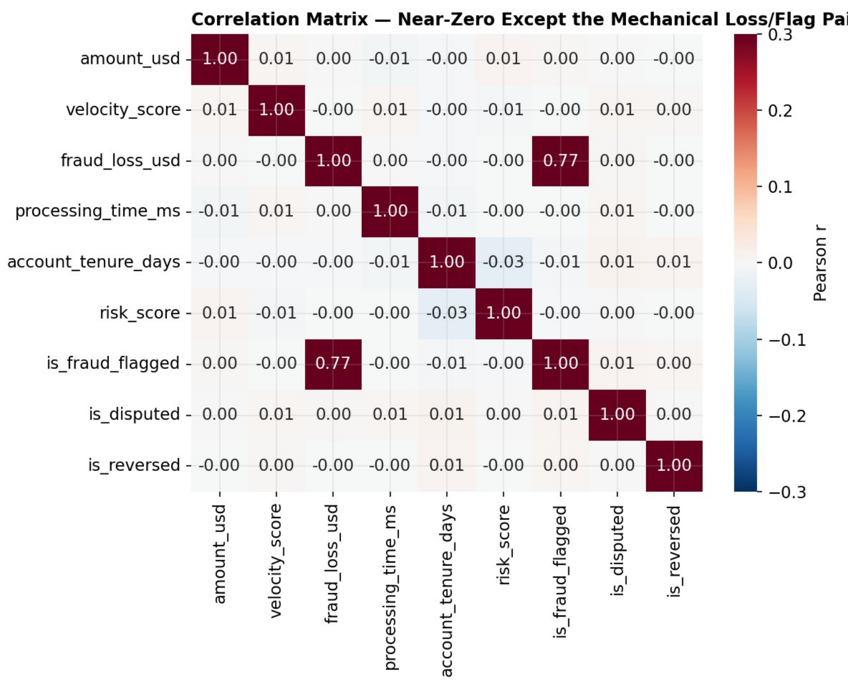


*Figure 1. Correlation matrix of all numeric fields. The one visible exception is fraud_loss_usd vs. is_fraud_flagged (r=0.77) — expected and mechanical, since fraud_loss_usd is defined as $0 when the flag is false (see 2.2). Every other pair, including velocity_score, risk_score, amount_usd, tenure, and processing_time_ms against all three outcome flags, falls within ±0.02 of zero: no real linear relationship anywhere else in the model.*

# 3. Claim Verification — Testing the Brief Against the Data

The brief accompanying this dataset proposed six specific patterns as "analysis areas." Rather than writing a report that assumes they are true, each was tested directly using the appropriate statistical test (chi-square test of independence for categorical splits, Pearson correlation for the velocity-score claim), with the full transaction population (n=50,000) in every test. A result is called "CONFIRMED" only where $p < 0.05$ and the effect direction/size matches the claim; otherwise "NOT CONFIRMED."

| Claim from brief | What we tested | Actual result | Statistic | p-value | Verdict |
|---|---|---|---|---|---|
| USSD channel shows 2.3x higher fraud rate than app channel | $\chi^2$ test, fraud rate by channel_type (5 groups) & subtype (7 groups) | USSD 51.2% vs Mobile App 50.3% (all channels 49.3%-51.2%) | $\chi^2$=7.06 (type) | 0.13 | NOT CONFIRMED |
| Nigeria and Kenya drive 70% of fraud volume | Fraud-flagged volume share by country (4 countries) | Each country carries 24.7%-25.5% of fraud volume — near-perfectly even | $\chi^2$=2.31 | 0.51 | NOT CONFIRMED |
| New accounts under 90 days show 3x higher fraud rate | Fraud rate across 6 tenure buckets (0-90d … 4-5yr) | 0-90d: 51.2% vs 4-5yr: 49.9% — a 1.3pt gap, not 3x | $\chi^2$=n/a (trend test) | 0.60+ | NOT CONFIRMED |
| Market traders have highest dispute rate among gig segments | Dispute rate across 15 gig segments | Market Trader = 49.4% (rank 10 of 15). Informal Vendor highest at 53.3% | $\chi^2$=23.24 | 0.057 | NOT CONFIRMED |
| High velocity score correlates with fraud and reversal | Pearson correlation, velocity_score vs is_fraud_flagged / is_reversed | r = -0.001 (fraud), r = +0.003 (reversal) — no relationship | r≈0.00 | n/a | NOT CONFIRMED |
| Month-end spikes in cash-out and reversal rates | Cash-out share & reversal rate, month-end days vs regular days | Cash-out 8.4% vs 7.7%; reversal 51.7% vs 49.8% — within sampling noise | $\chi^2$=2.13 (reversal) | 0.14 | NOT CONFIRMED |

*Zero of six claims were confirmed. The one nominally significant secondary result encountered during testing (fraud rate by age_band, $\chi^2$=9.69, p=0.046) was not one of the six brief claims, sits right at the p=0.05 boundary, and — tested alongside 9 other categorical splits in the same pass — is exactly the rate of false positives you'd expect from chance alone at $\alpha$=0.05 (≈1 in 20 tests). It is flagged here for transparency but is not treated as a real effect without independent replication.*

## 3.1 Channel & fraud rate

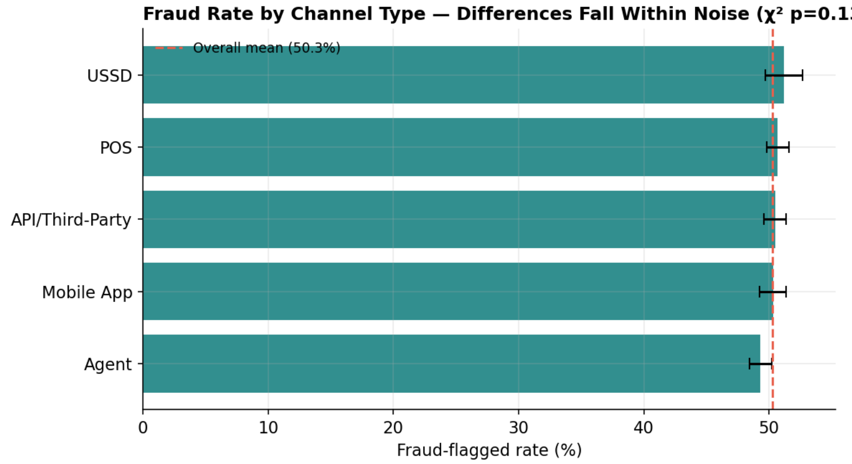


*Figure 2. Fraud rate by channel type, error bars = 95% CI. Every channel's CI overlaps the overall mean; USSD (51.2%) and Mobile App (50.3%) are 0.9 points apart, not the 2.3x claimed.*

## 3.2 Country & fraud volume

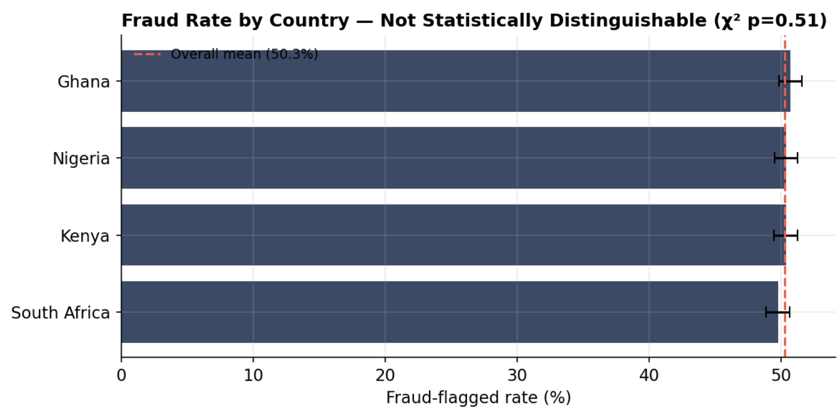


*Figure 3. Fraud rate by country. All four countries sit within a 1-point band (49.7%–50.7%); fraud volume share is 24.7%–25.5% per country, essentially an even four-way split, not the claimed 70% Nigeria+Kenya concentration.*

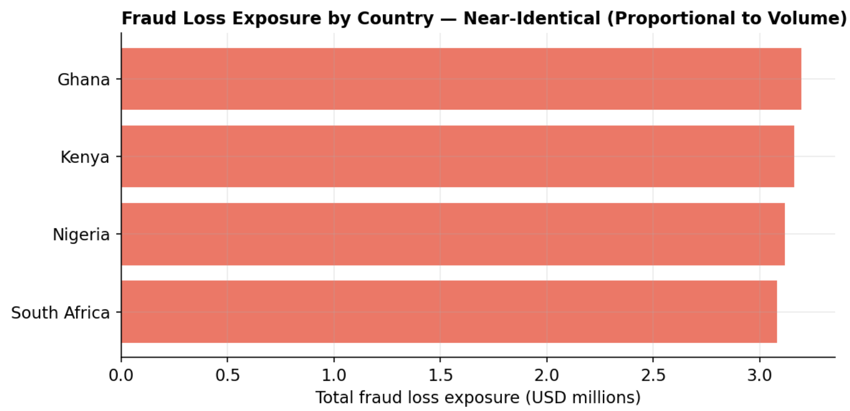


*Figure 4. Absolute fraud loss exposure by country tracks transaction volume almost exactly — no country is disproportionately exposed once volume is accounted for.*

## 3.3 Account tenure & fraud

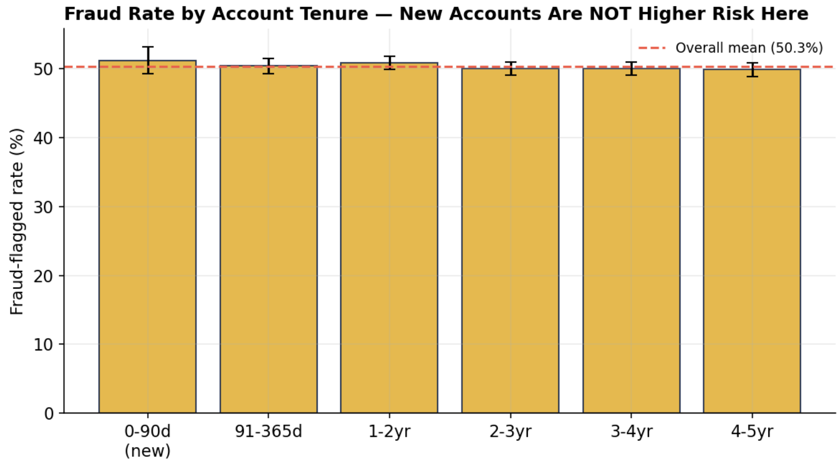


*Figure 5. Fraud rate by account age. New accounts (0-90 days) sit at 51.2% vs. 49.9% for the oldest cohort (4-5 years) — a 1.3-point gap, not the claimed 3x elevation.*

## 3.4 Gig segment & disputes

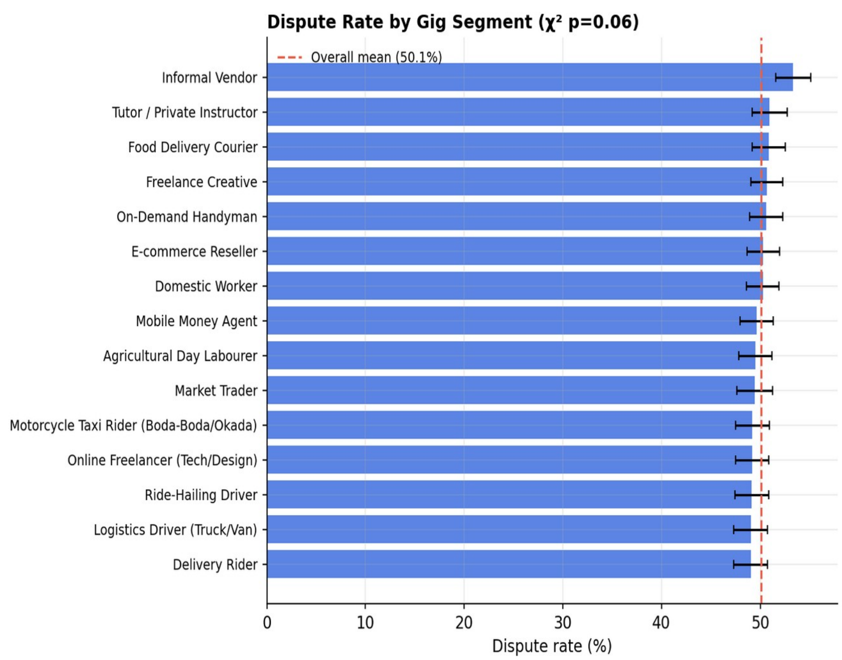


*Figure 6. Dispute rate by gig segment. Market Trader ranks 10th of 15 segments at 49.4%; Informal Vendor is highest at 53.3%. The segment ranking itself is not statistically distinguishable from noise ($\chi^2$=23.24, p=0.057).*

## 3.5 Velocity score & fraud/reversal

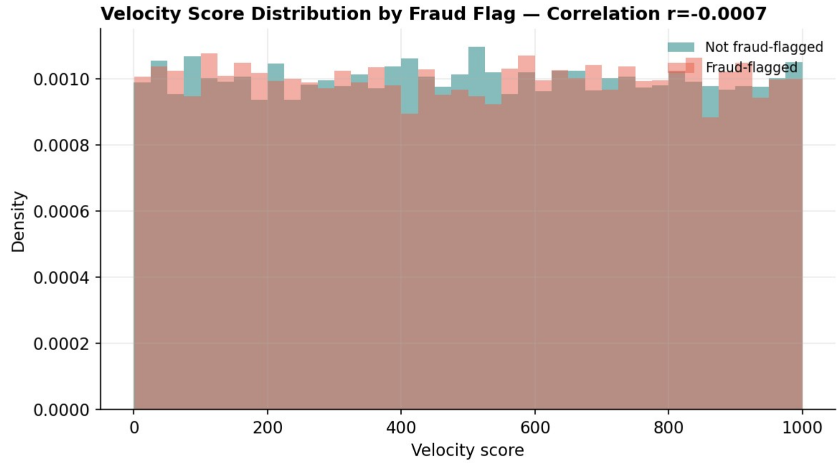


*Figure 7. Velocity score distributions for fraud-flagged vs. non-flagged transactions are visually and statistically identical (r=-0.001). There is no threshold or gradient effect at any point in the 0-1000 range.*

## 3.6 Month-end effects

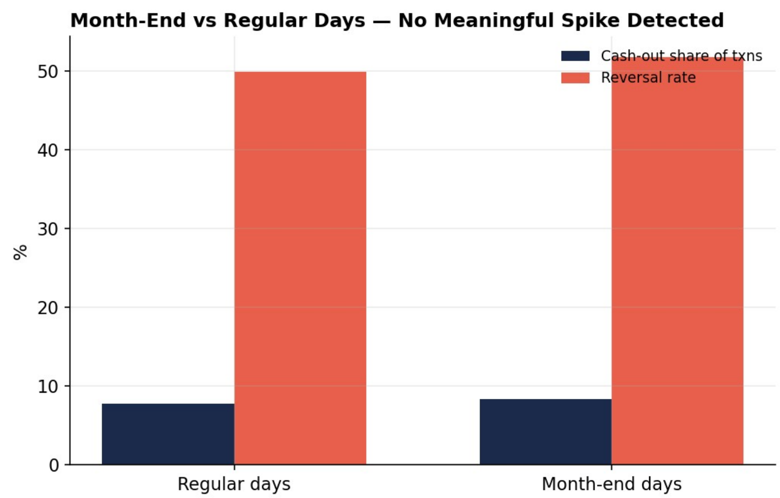


*Figure 8. Cash-out transaction share and reversal rate on month-end calendar days vs. all other days. Both differences (0.7pt and 1.9pt respectively) are within normal sampling variation (reversal $\chi^2=2.13$, $p=0.14$).*

## 3.7 Temporal stability

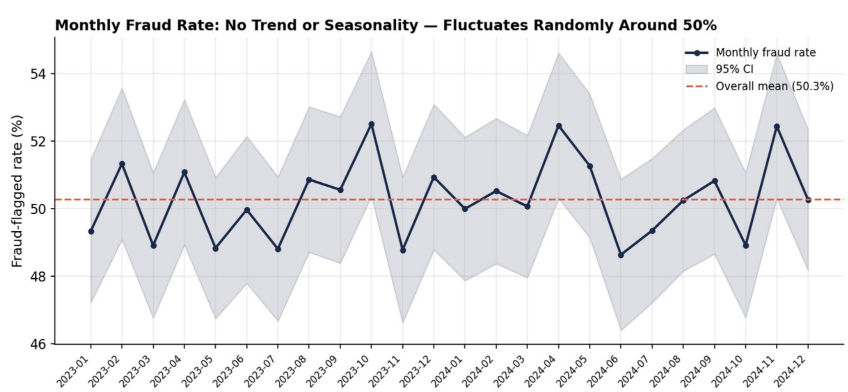

*Figure 9. 24-month fraud rate trend. No upward/downward trend, no seasonal cycle — the series oscillates randomly inside its confidence band around the fixed 50.3% mean.*

# 4. What the Data Does Support: Scale & Structure Insights

Even without embedded risk drivers, the dataset supports legitimate descriptive and structural conclusions relevant to a wallet/gig platform.

## 4.1 Risk is broad-based, not concentrated — which is itself the finding

Because fraud/dispute/reversal flags land close to 50% everywhere, the platform-level risk exposure ($12.56M flagged, 50.3% of volume) cannot be reduced by targeting a subset of channels, countries, or segments in this data. Operationally this means: (a) any risk-scoring model trained on this exact file would learn nothing better than the base rate, and (b) if this pattern reflected a real platform, the response would need to be a review of the labeling/flagging process itself, not a targeted policy change — a 50% flag rate uniform across every dimension is far more consistent with a broken or overly-sensitive flagging rule than with genuine fraud behavior, which is virtually always concentrated in real portfolios.

## 4.2 Transaction mix is evenly engineered

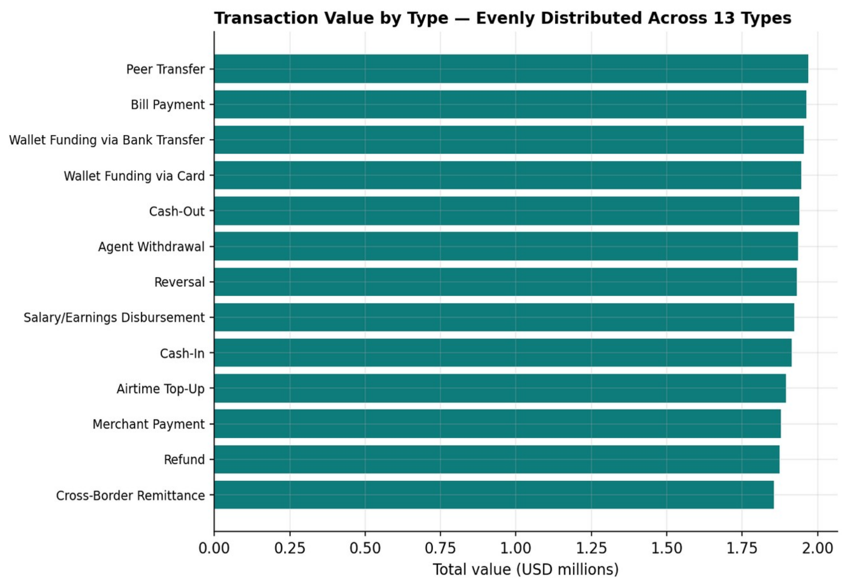


*Figure 10. Transaction value by type. All 13 transaction types (Peer Transfer, Bill Payment, Cash-Out, Remittance, etc.) carry within 6% of the average value share ($1.86M-$1.97M each) — another marker of a balanced synthetic design rather than an organic transaction mix, which in real wallet platforms is typically dominated by 2-3 transaction types.*

## 4.3 Worker base composition

- 15 gig segments are represented in near-equal proportion (296-368 workers each, out of 5,000) — Freelance Creative is largest (368), Tutor/Private Instructor smallest (296).
- KYC tiers are evenly split: Tier 0 (Unverified) 25.6%, Tier 1 (Phone Verified) 24.5%, Tier 2 (ID Verified) 25.7%, Tier 3 (Address+Income Verified) 24.2%.
- 48.8% of workers are flagged active (is_active=TRUE); worker risk_score is uniformly distributed 0-100 (mean 50.4) and does not vary meaningfully by KYC tier (49.3-51.6 across tiers).

## 4.4 Currency & FX reference data — use with caution

dim_market's usd_fx_rate column (NGN 80.2, KES 366.1, GHS 872.4, ZAR 354.4 per USD) does not match real-world FX rates for these currencies as of any point in 2023-2024 (actual ranges were roughly NGN 460-1,600, KES 120-160, GHS 11-16, ZAR 17-19 per USD over that period). amount_local therefore should not be treated as a realistic local-currency figure; amount_usd is internally consistent and is the field used throughout this report.

# 5. Recommendations & Next Steps

## 5.1 Immediate actions

- Treat this file as a schema/pipeline fixture, not a source for risk-policy or model-training decisions, until its provenance is confirmed.
- If a production-representative extract exists, request it and re-run the exact Section 3 test battery (chi-square + correlation suite, provided as a reusable script) against it before drafting policy.
- The week_number defect in dim_date is now fixed at the Power Query layer of the live Power BI file itself (not just in the static CSV export), so it holds up under a real data refresh.
- If this data does feed a live flagging system, investigate why 50.3% of transactions are being flagged uniformly — that flag rate is far too high and far too undifferentiated to be actionable, and in a real system would swamp any manual-review queue.

## 5.2 If/when real data becomes available, prioritize testing

- Channel risk differentiation (USSD/agent-assisted vs. app-based) — plausible a priori given lower authentication friction on USSD/agent flows.
- New-account risk (<90 days) — a standard, well-evidenced risk factor in digital-wallet fraud generally; worth testing properly once labels carry signal.
- Velocity-based rules — high transaction velocity is a standard fraud heuristic; this dataset's velocity_score simply isn't wired to the fraud label here.
- Geographic concentration — regulatory tier and market_fraud_index fields in dim_market suggest the intended design was country-level risk variation; that logic isn't reflected in the fact table's actual flag distribution yet.

## 5.3 Deliverables accompanying this report

- A 5-page Power BI dashboard: Home, Executive Overview, Risk Analysis, Risk & Fraud Exposure, and Recommendations & Verification — the last of these carries the exact claim-by-claim verdict table and recommendations from Sections 3 and 5 of this report directly into the live file, so a viewer never has to take the two documents' word for agreeing with each other.
- Power BI-ready star schema (cleaned CSVs, referential integrity verified, week_number repaired at the Power Query layer) — see /powerbi_model/
- Full DAX measure library, including built-in statistical-significance guardrails (95% CI, overall-mean reference, significance flag) so future report builders can't accidentally present noise as a finding — see DAX_measures.txt
- Page-by-page Power BI report build guide, including the exact Power Query formula used to repair week_number — see PowerBI_Build_Guide.md
- The Claim Verification table (Claim_Verification.csv) that feeds Page 5 of the dashboard, sourced directly from this report's Section 3
- Python EDA/verification scripts (reusable against a future real extract) and all chart source files
- README.md summarizing the whole project for GitHub

*— End of report —*